OFFICIAL COLLEGE PROJECT ABSTRACT · PROJECT 33 OF 55

REMAINING USEFUL LIFE PREDICTION OF AIRCRAFT COMPONENTS USING MACHINE LEARNING

Domain: Aerospace Prognostics & NASA C-MAPSS AI | **Module:** project_33_aircraft_rul

OFFICIAL ABSTRACT

Remaining Useful Life Prediction of Aircraft Components Using Machine Learning is a comprehensive and crucial project aimed at addressing unexpected equipment failures, costly downtime, and safety risks in modern aerospace environments. As aviation complexity accelerates, airlines and manufacturers increasingly rely on interconnected systems that continuously generate complex operational metrics, making them prime targets for sudden mechanical breakdowns if component degradation is not proactively identified. Traditional maintenance methods often rely on scheduled or reactive servicing, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly developing, complex mechanical anomalies and subtle wear and tear. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time aircraft health monitoring and remaining useful life (RUL) prediction. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive flight sensor streams, such as fluctuating engine temperatures, vibration patterns, pressure variations, and operational speed deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and forecasting algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine operation for each distinct aircraft component, the model establishes a dynamic baseline that adapts to changing flight load requirements over time. The proposed model is rigorously trained, validated, and tested on robust datasets encompassing diverse aerospace sensor logs, operational flight cycles, and simulated degradation scenarios. This enables the framework to accurately distinguish between healthy component operation and critical anomalies—such as

turbine blade degradation, hydraulic system wear, and structural imbalances—with high precision and minimal latency. Experimental results and comprehensive performance evaluations demonstrate that the seamless integration of AI-driven feature extraction and machine learning models provides significantly improved classification and regression performance, resilience, and adaptability compared to traditional legacy maintenance approaches.

📌 Project Summary & System Specifications

Project Title	REMAINING USEFUL LIFE PREDICTION OF AIRCRAFT COMPONENTS USING MACHINE LEARNING
Specialization / Category	Aerospace Prognostics & NASA C-MAPSS AI
Target Technology Stack	Python (Flask), PyTorch / TensorFlow / Scikit-Learn, Modern Web UI
Default Backend Port	`:5033`
Local Implementation	`project_33_aircraft_rul`